Exp. No: 9 Implementation of Magnetic-Field Boundary Conditions between Two Dielectric Media and Determination of Relative Permeability

Aim:

To implement magnetic-field boundary conditions between two media and determine the relative permeability of the unknown medium.

Objectives:

1. To resolve magnetic-field intensity into normal and tangential components at a material interface.

2. To apply the tangential H-field and normal B-flux-density boundary conditions.

3. To determine the relative permeability of an unknown medium from specified boundary data.

4. To verify the magnetic boundary conditions numerically using MATLAB.

Test Case (i) – Magnetic Boundary Conditions and Determination of the Relative Permeability

Apply the boundary conditions H2t = H1t and B2n = B1n at the interface, determine the relative permeability of medium 2 from the measured normal H-field and compare it with the reference value.

```
clc;

clear;

close all;


% Magnetic-Field Boundary Conditions between Two Media

mu0 = 4*pi*1e-7;   % H/m


% Input data

mur1 = input('Enter relative permeability of medium 1: ');

H1t = input('Enter tangential H-field H1t (A/m): ');

B1n = input('Enter normal flux density B1n (T): ');

H2n_meas = input('Enter measured/assigned normal H-field H2n (A/m): ');

mur2_ref = input('Enter reference relative permeability of medium 2: ');
```

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```
% Permeability of the two media

mu1 = mu0*mur1;

mu2_ref = mu0*mur2_ref;


% Boundary conditions for zero surface current

H2t = H1t;

B2n = B1n;


% Determine relative permeability from measured H2n

mur2_calc = B2n/(mu0*H2n_meas);


% Theoretical normal H-field using reference permeability

H2n_theory = B2n/mu2_ref;


% Percentage deviation

dev = abs(mur2_calc-mur2_ref)/abs(mur2_ref)*100;


% Display results

fprintf('\n--- Magnetic Boundary Condition Results ---\n');

fprintf('H2t (theoretical)    = %.6f A/m\n',H2t);

fprintf('B2n (theoretical)    = %.6e T\n',B2n);

fprintf('H2n (theoretical)    = %.6f A/m\n',H2n_theory);

fprintf('Calculated mur2      = %.6f\n',mur2_calc);

fprintf('Reference mur2       = %.6f\n',mur2_ref);

fprintf('Percentage deviation = %.4f %%\n',dev);
```

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```
% Parameter sweep: effect of relative permeability on normal H-field

mur = linspace(1,10,100);

Hn = B1n./(mu0*mur);


figure;

plot(mur,Hn,'LineWidth',2);

grid on;

xlabel('Relative permeability, \mu_r');

ylabel('Normal magnetic field H_n (A/m)');

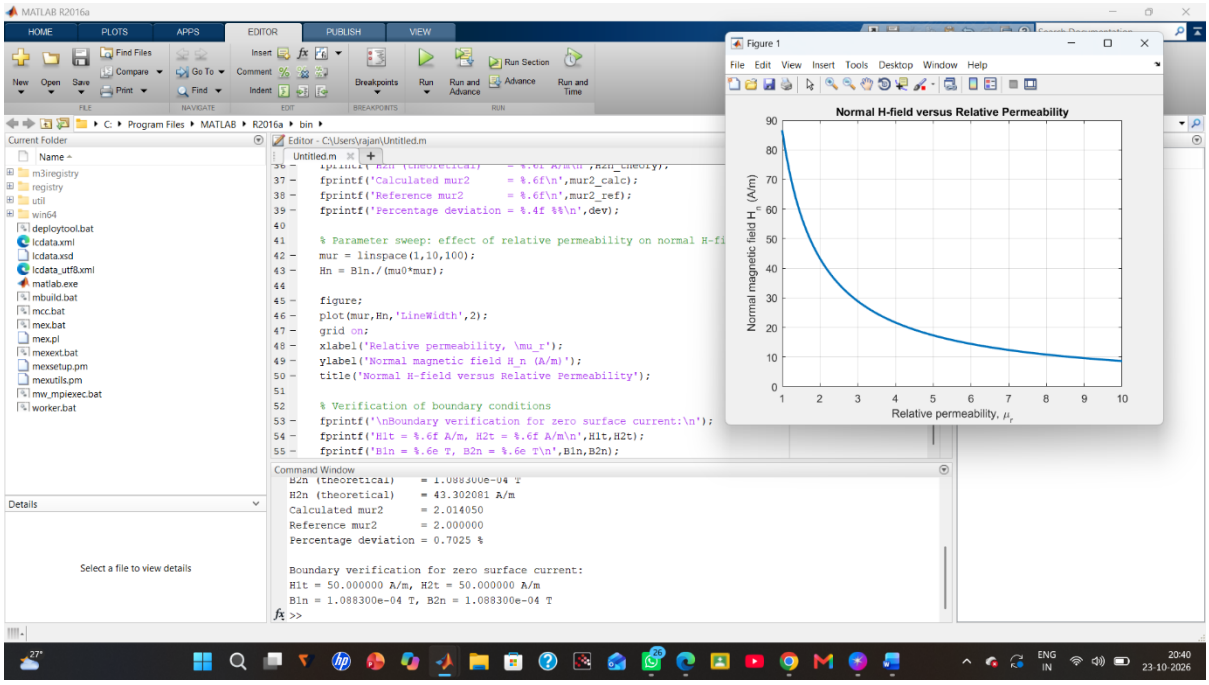
title('Normal H-field versus Relative Permeability');


% Verification of boundary conditions

fprintf('\nBoundary verification for zero surface current:\n');

fprintf('H1t = %.6f A/m, H2t = %.6f A/m\n',H1t,H2t);

fprintf('B1n = %.6e T, B2n = %.6e T\n',B1n,B2n);
```



Test Case (ii) – Field Components and Refraction Angle at the Interface of Two Magnetic Media

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Determine the field components, normal flux density and refraction angle in medium 2 for a specified incident field and specified relative permeabilities.

```
clc; clear;

mu0  = 4*pi*1e-7;

H1   = input('Enter magnitude of H1 in medium 1 (A/m): ');

th1  = input('Enter angle of H1 with the normal (deg): ');

mur1 = input('Enter relative permeability of medium 1: ');

mur2 = input('Enter relative permeability of medium 2: ');

H1n = H1*cosd(th1);              % normal component in medium 1

H1t = H1*sind(th1);

      % tangential component in medium 1

B1n = mu0*mur1*H1n;              % normal flux density in medium 1

H2t = H1t;                       % tangential H is continuous (Ks = 0)

B2n = B1n;                       % normal B is continuous

H2n = B2n/(mu0*mur2);            % normal H in medium 2

H2  = sqrt(H2n^2 + H2t^2);

th2 = atand(H2t/H2n);

fprintf('H1n = %.2f A/m, H1t = %.2f A/m\n', H1n, H1t);

fprintf('B1n = %.4e T\n', B1n);

fprintf('H2t = %.2f A/m, B2n = %.4e T\n', H2t, B2n);

fprintf('H2n = %.2f A/m, |H2| = %.2f A/m\n', H2n, H2);

fprintf('Refraction angle theta2 = %.2f deg\n', th2);

figure; plot([0 0],[-80 80],'k','LineWidth',2); hold on;

quiver(-H1n,-H1t,H1n,H1t,0,'b','LineWidth',2);

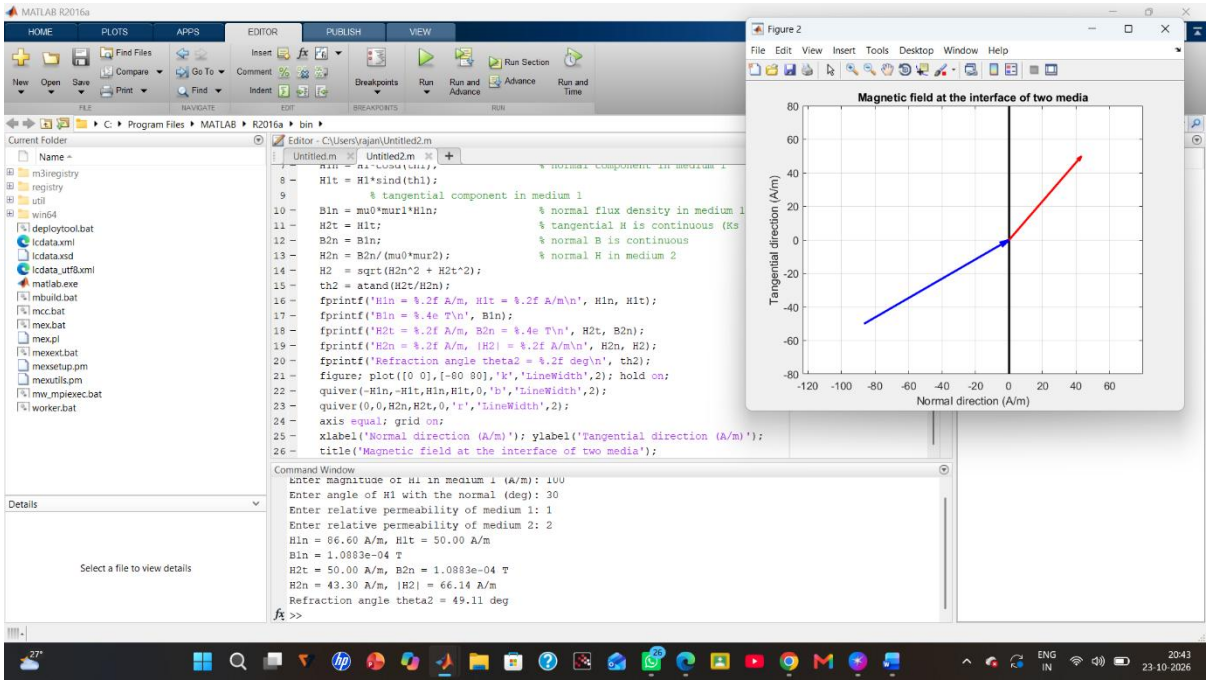
quiver(0,0,H2n,H2t,0,'r','LineWidth',2);

axis equal; grid on;

xlabel('Normal direction (A/m)'); ylabel('Tangential direction (A/m)');
```

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title('Magnetic field at the interface of two media');



Test Case (iii) – Variation of the Refraction Angle with the Angle of Incidence for Different Permeabilities

Plot the angle of the magnetic field in medium 2 against its angle in medium 1 for three values of μr2 with μr1 = 1.

clc; clear;

th1 = linspace(0,89,179);

mur1 = 1;

mur2 = [2 5 10];

figure; hold on;

for k = 1:3

th2 = atand((mur2(k)/mur1)*tand(th1)); % tan(th2) = (mur2/mur1)*tan(th1)

plot(th1,th2,'LineWidth',2);

end

plot(th1,th1,'k--'); % mur2 = mur1 (no change of direction)

grid on;

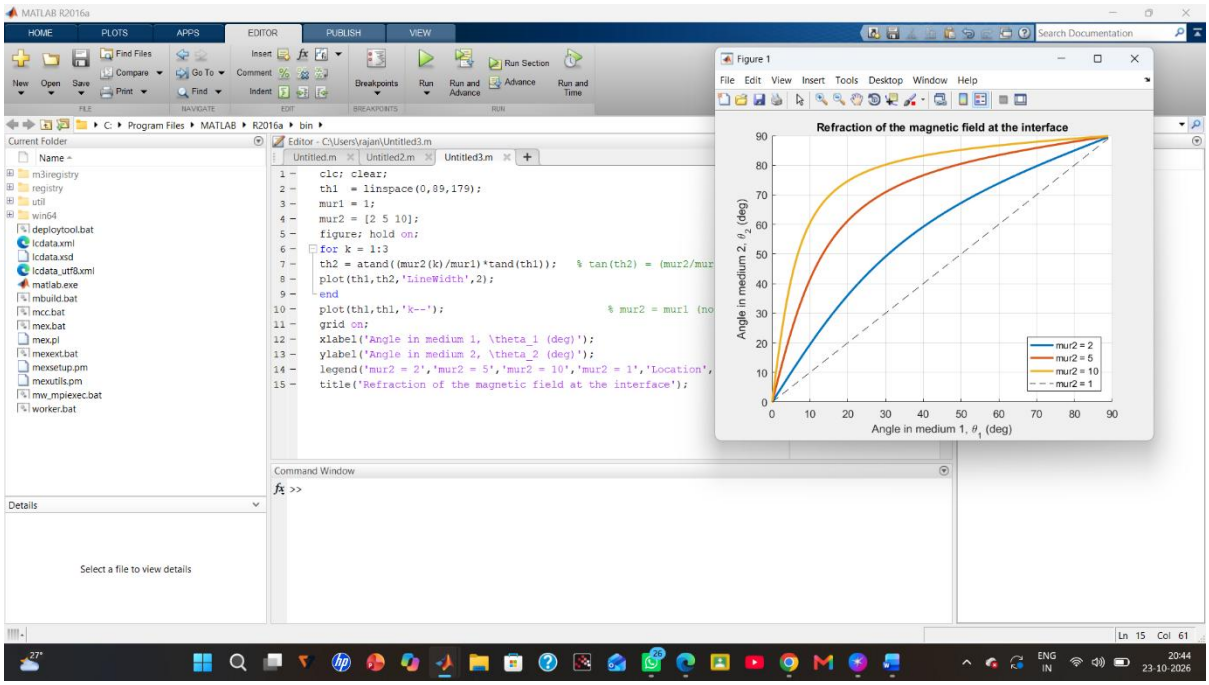
xlabel('Angle in medium 1, \theta_1 (deg)');

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ylabel('Angle in medium 2, \theta_2 (deg)');

legend('mur2 = 2','mur2 = 5','mur2 = 10','mur2 = 1','Location','southeast');

title('Refraction of the magnetic field at the interface');



Test Case (iv) – Effect of the Relative Permeability on the Field Components in Medium 2
Keep the incident field fixed and vary μr2 to show which components are continuous across the boundary.

clc; clear;

mu0 = 4*pi*1e-7;

H1 = 100; th1 = 30; mur1 = 1;

H1n = H1*cosd(th1); H1t = H1*sind(th1);

B1n = mu0*mur1*H1n;

mur2 = linspace(1,10,100);

H2n = B1n./(mu0*mur2); % normal H in medium 2

H2t = H1t*ones(size(mur2)); % tangential H (continuous)

H2 = sqrt(H2n.^2 + H2t.^2);

B2n = mu0*mur2.*H2n; % normal B (continuous)

subplot(1,2,1);

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plot(mur2,H2n,mur2,H2t,mur2,H2,'LineWidth',2); grid on;

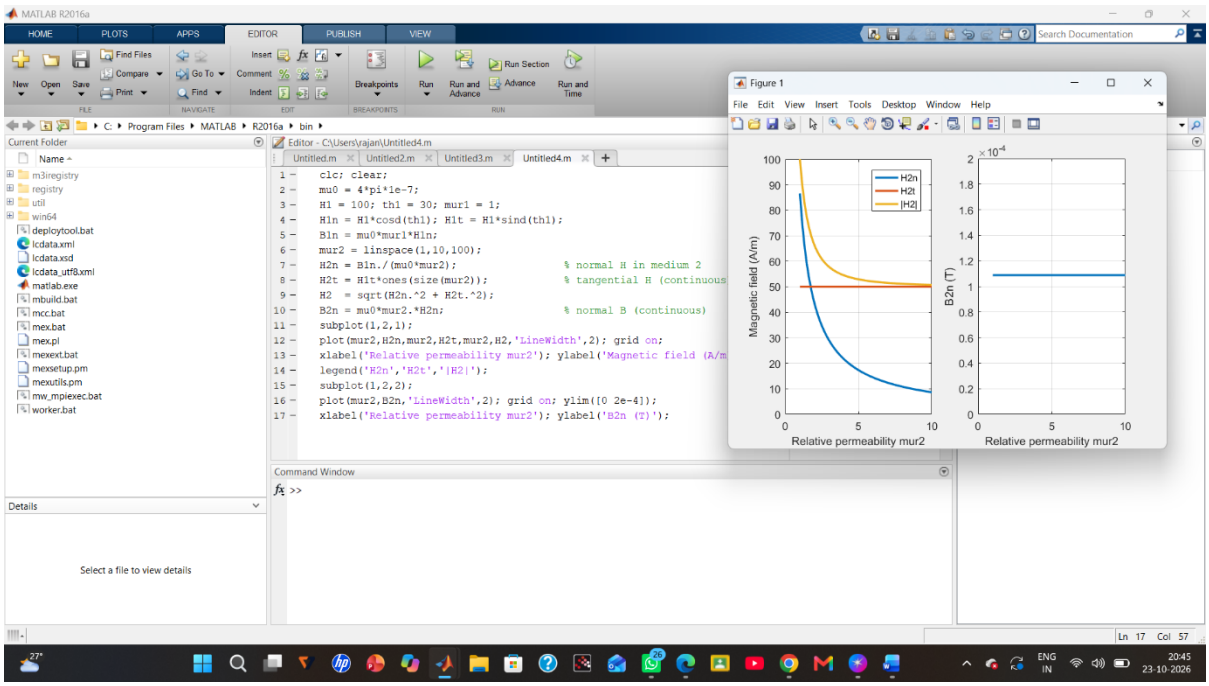
xlabel('Relative permeability mur2'); ylabel('Magnetic field (A/m)');

legend('H2n','H2t','|H2|');

subplot(1,2,2);

plot(mur2,B2n,'LineWidth',2); grid on; ylim([0 2e-4]);

xlabel('Relative permeability mur2'); ylabel('B2n (T)');



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Identify magnetic media and boundary orientation	Identify magnetic media and boundary orientation	Correctly identifies permeability, normal/tangential components and surface current assumptions.	Setup is correct with minor omissions.	Needs help identifying components or assumptions	Uses incorrect interface definitions.	25	2.5	
2	Apply magnetic boundary relations	Apply magnetic boundary relations	Correctly applies continuity of normal B and the tangential H relation.	Relations are applied with minor errors.	Partly correct application requires guidance.	Boundary relations are incorrect.	25	2.5	
3	Determine	Determine	Calculates	Results are	Validation is	Results are	30	3	

	permeabili ty and validate fields	permeabili ty and validate fields	required quantities and verifies the boundary relations consistently.	mostly correct with minor discrepa ncies.	incomplete.	invalid or unverified.			
4	Explain and document findings	Explain and document findings	Explains magnetic-field behaviour and records assumptions, units and conclusion.	Docume ntation is clear with minor omissio ns.	Interpretation or documentatio n is incomplete.	Cannot explain or report results.	20	2	
Tota l							100	10	